

Supplementary Material

1 PARTICIPANT AND BEHAVIOURAL DETAILS

Table S1. Clinical motor severity by medication state. MDS-UPDRS Part III ratings were available for all 18 participants. Values are reported as mean $\pm$ SD, median [range], or n/N.

Measure	OFF medication	ON medication	ON-OFF
MDS-UPDRS Part III total, mean $\pm$ SD	37.3 $\pm$ 17.3	28.0 $\pm$ 15.0	−9.3 $\pm$ 4.2
MDS-UPDRS Part III total, median [range]	35.0 [10, 75]	22.0 [7, 61]	−9.0 [−17, −3]
Improvement from OFF, %, mean $\pm$ SD	—	26.6 $\pm$ 9.1	—
Minutes since last dose, median [range]	931.5 [720, 3006]	71.0 [36, 102]	—
H&Y stage, median [range]	2 [1, 4]	2 [0, 3]	—
Dyskinesias present, n/N	1/18	3/18	—
Dyskinesias interfered with ratings, n/N	1/18	0/18	—

Table S2. MDS-UPDRS Part III domain scores by medication state. Domain groupings are descriptive and are defined by the item numbers listed in the table. Values are reported as mean $\pm$ SD.

Domain	Items included	OFF mean $\pm$ SD	ON mean $\pm$ SD	ON-OFF mean $\pm$ SD
Axial/signs	3.1, 3.2, 3.9–3.14	8.4 $\pm$ 4.6	7.3 $\pm$ 4.5	−1.2 $\pm$ 1.3
Rigidity	3.3A–3.3E	6.5 $\pm$ 3.9	5.8 $\pm$ 3.6	−0.7 $\pm$ 1.0
Limb bradykinesia	3.4A–3.8B	15.1 $\pm$ 7.6	11.3 $\pm$ 7.8	−3.8 $\pm$ 3.3
Tremor	3.15A–3.18	7.2 $\pm$ 5.3	3.6 $\pm$ 3.1	−3.6 $\pm$ 3.3

2 BEHAVIOURAL QUALITY CONTROL AND REACTION-TIME SUMMARIES

This section reports clinical summaries and behavioural quality-control information that support the main text without interrupting the main Results narrative.

Reaction times were modestly faster in the ON-medication condition than in the OFF-medication condition. This pattern was visible both in the trial-wise RT distributions, where the ON distribution was slightly shifted toward shorter latencies, and in the subject-paired means, where most participants showed lower mean RT ON medication. The paired subject-level test showed a mean ON-OFF difference of −13.6 ms, 95% CI [−27.5, 0.3], $t(16) = -2.08$, $p = .054$, indicating a small medication-related speeding effect that approached but did not reach conventional significance. By comparison, temporal effects within the task were limited: RT was only slightly slower in run 2 than run 1, block position showed a mild increase in RT consistent with weak fatigue or time-on-task effects, and trial position within blocks showed no clear monotonic practice or drift pattern. Overall, the behavioural results suggest a small medication-associated reduction in RT, with practice, fatigue, and within-task drift contributing only modestly to RT variation.

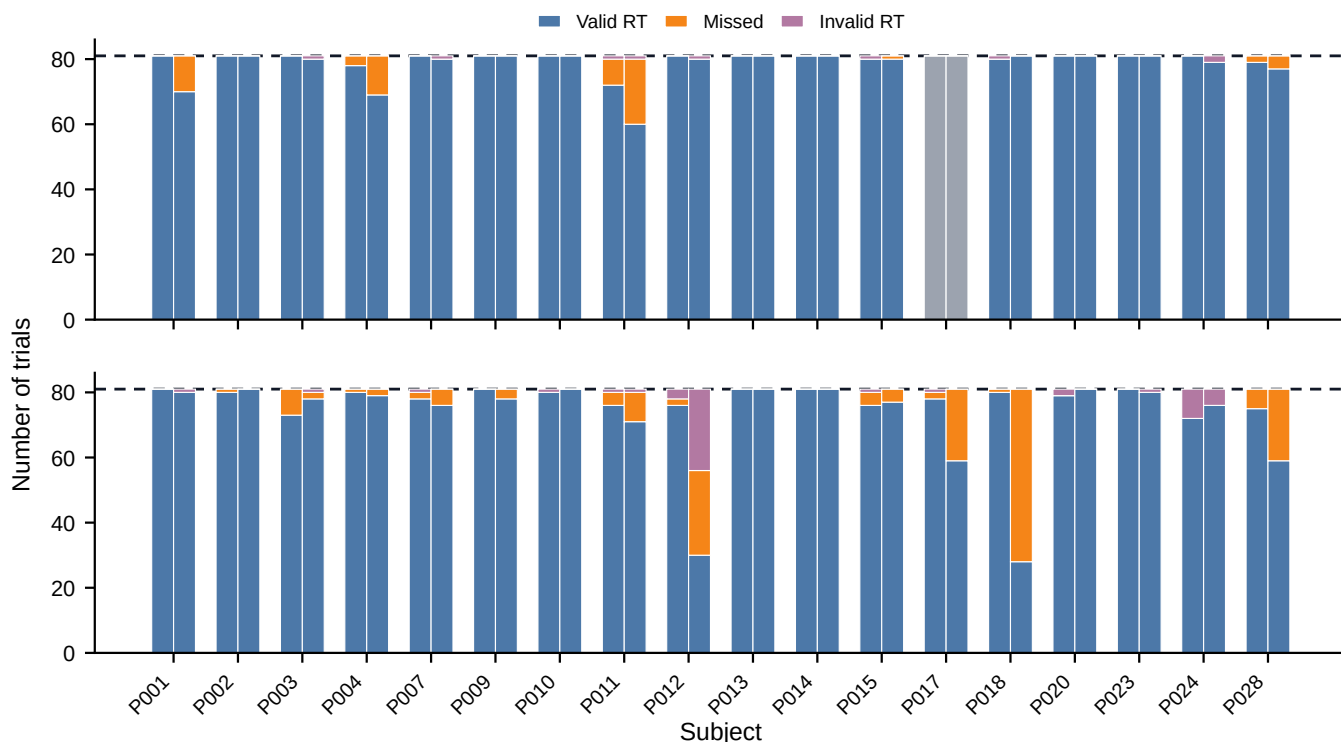


Figure S1: Reaction-time quality-control summary by participant, medication session, and run. Stacked bars show valid, missed, and invalid go trials; top and bottom panels show OFF- and ON-medication sessions, respectively. Within each participant, the left and right bars correspond to runs 1 and 2. Each run contained 90 trials, including 9 catch trials not shown here and 81 go trials, marked by the dashed horizontal line. Valid RT trials were go trials with finite positive $1/RT$ values; missed trials were go trials flagged as no-squeeze events. Gray bars indicate data without corresponding fMRI data.

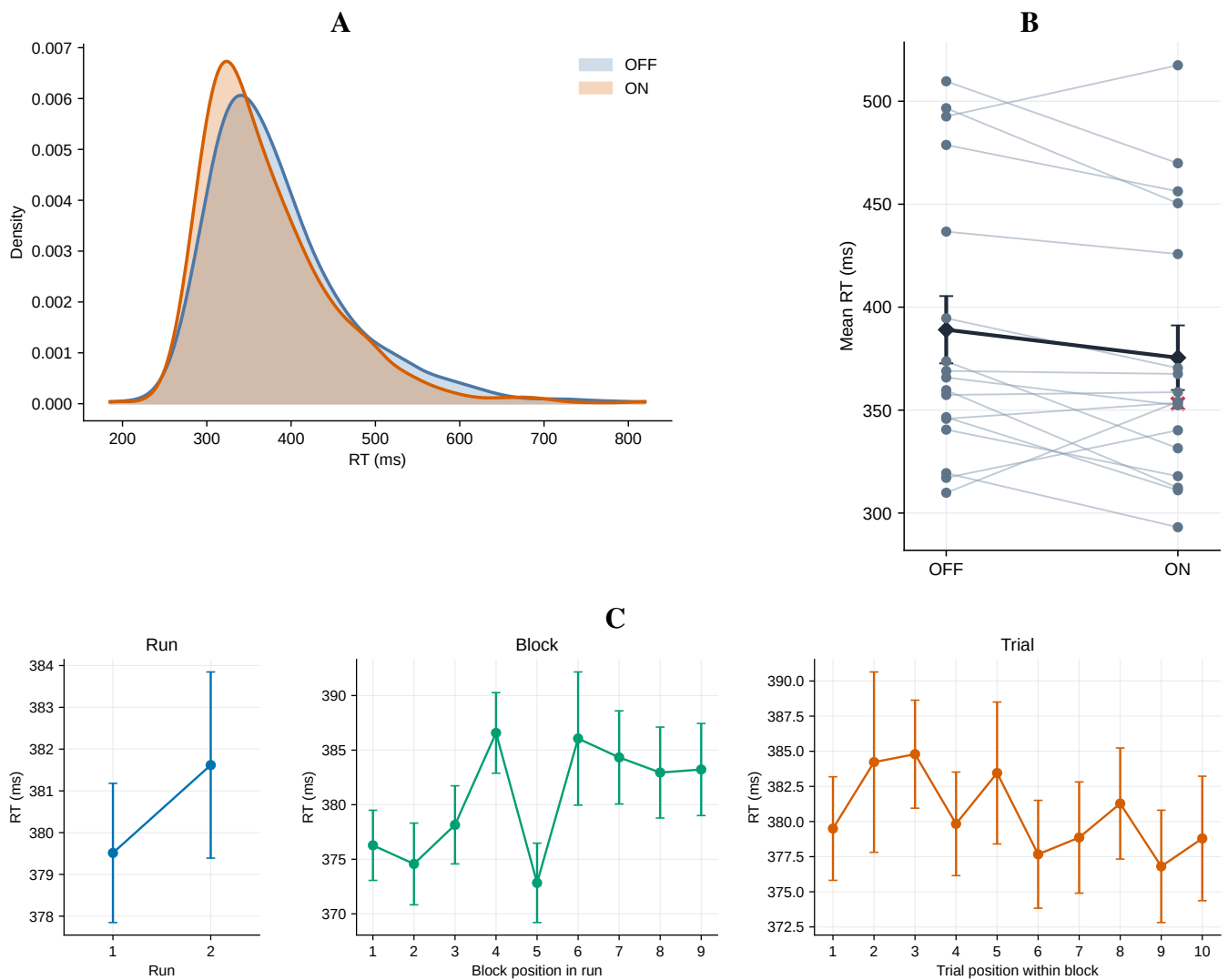


Figure S2: Supplementary behavioural reaction-time analyses. **A**, Trial-wise reaction-time distributions for OFF- and ON-medication sessions. **B**, Participant-level mean reaction times in the OFF- and ON-medication sessions. **C**, Mean reaction time by run, block position, and trial position within block, showing only modest temporal variation within the task.

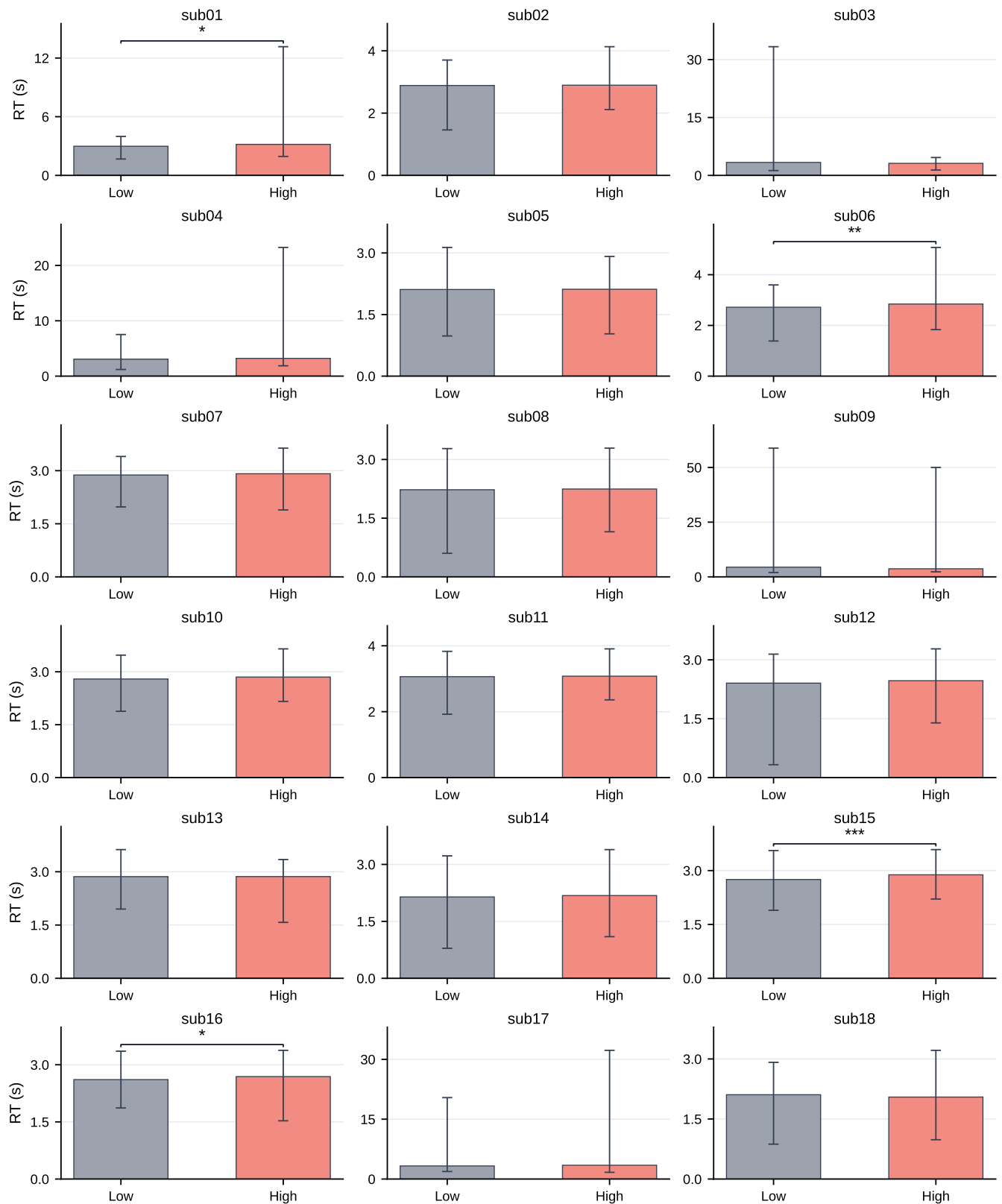


Figure S3: Reward effects on reaction time by subject. Bars show mean reaction time for low- and high-reward trials within each subject. Error bars indicate the full observed RT range from minimum to maximum. Asterisks mark significant high- versus low-reward differences from Welch two-sample tests across trials within subject ($p < 0.05$, $*p < 0.01$, $**p < 0.001$).

3 TASK-ACTIVATION REFERENCE MAPS AND ACTIVATION-MAP COMPARISONS

Spatial distribution of the standard GLM from the group-level mixed-effects analysis. The thresholded map indicates widespread effects across posterior and lateral cortical regions, with the strongest clusters shown in yellow-orange.

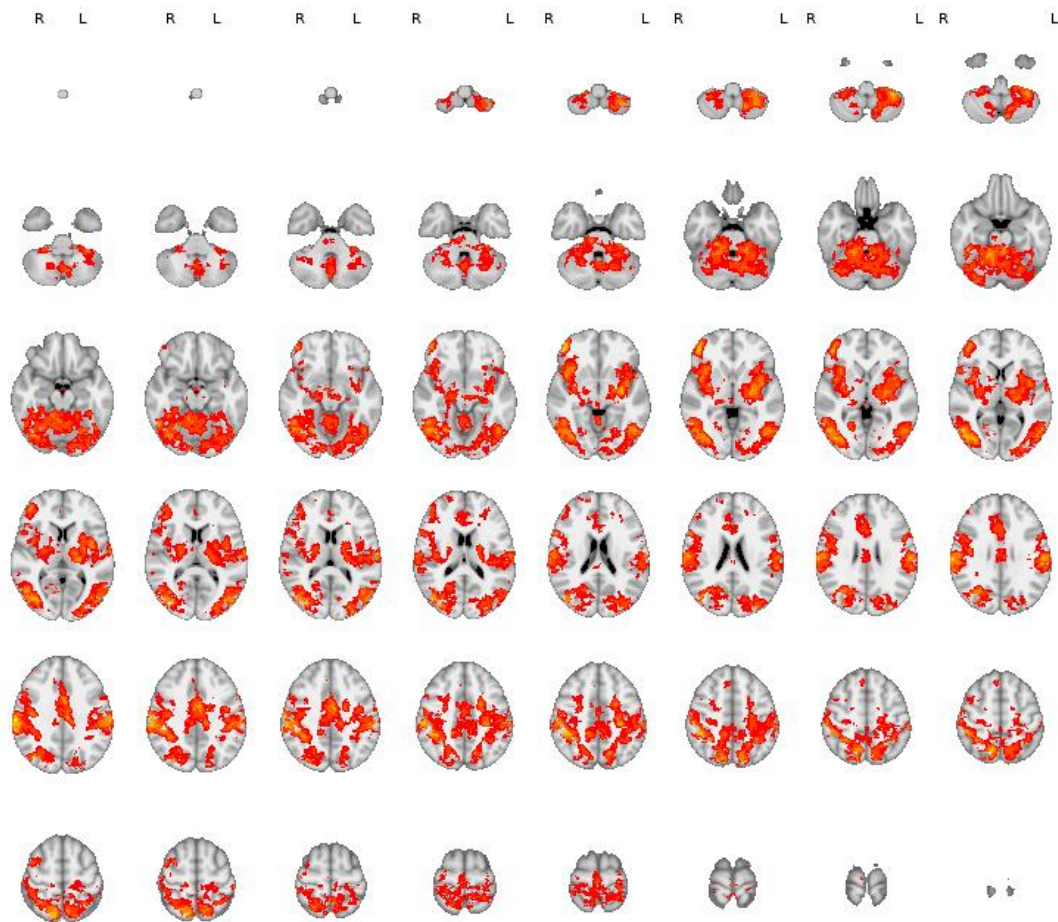


Figure S4: Group-level mixed-effects map. Axial brain slices show thresholded positive z-statistic values overlaid on the anatomical reference image. Warm colors indicate suprathreshold effects from the FEAT thresholded z-statistic image, with brighter colors reflecting higher z-values. The map is displayed at a cluster-forming threshold of $Z \geq 3.1$; R labels indicate image orientation.

Spatial similarity between activation-derived maps and the optimized vigour network is summarized below.

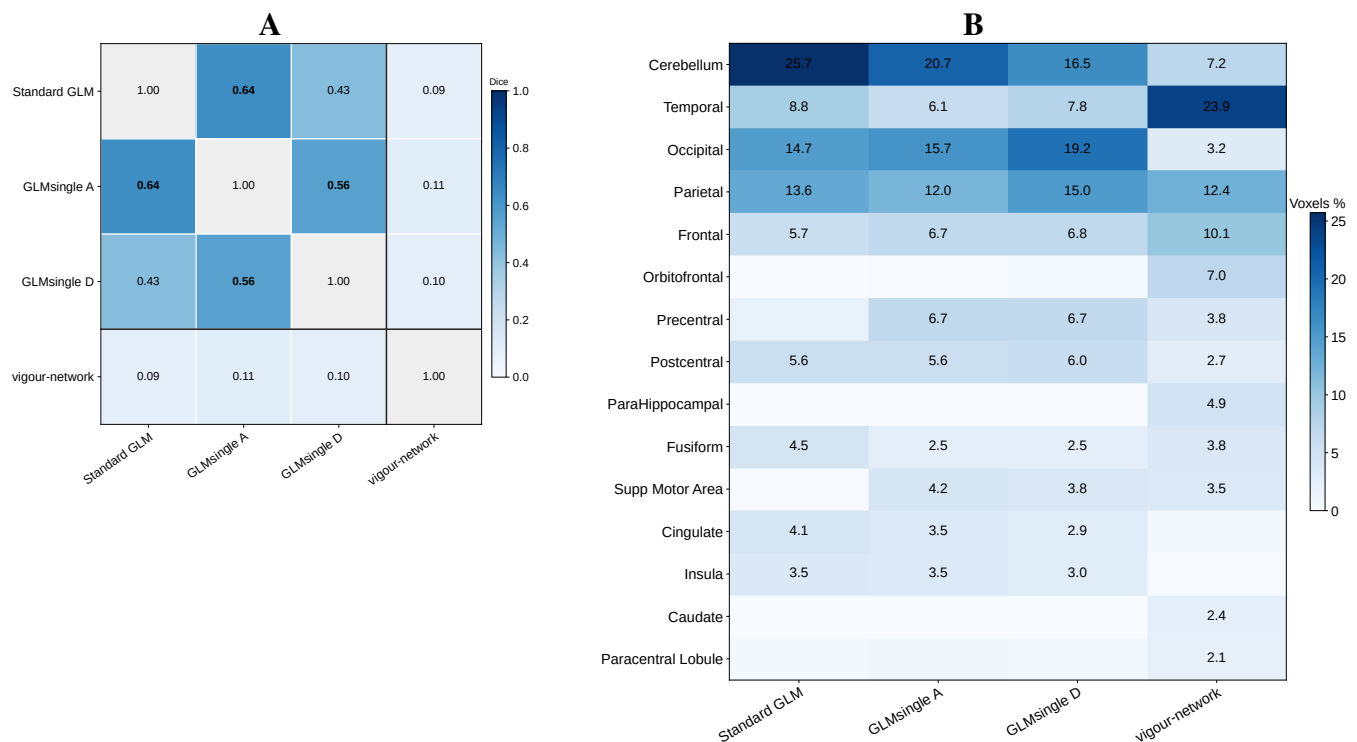


Figure S5: Spatial similarity between activation-derived maps and the optimized vigour network. **A**, Dice coefficient matrix comparing the standard GLM map, two GLMsingl activation maps, and the optimized vigour network. **B**, Regional composition of each map, expressed as the percentage of suprathreshold voxels assigned to each anatomical group. The vigour network showed low spatial overlap with activation-derived maps and a distinct regional profile.

4 ANATOMICAL CHARACTERIZATION OF THE VIGOUR NETWORK

4.1 Threshold Robustness Analysis

We tested whether the final network definition depended strongly on the voxel-weight threshold. The threshold was varied across the voxel-weight distribution at p80, p85, p90, p95, and p97.5 (Figure S6). To summarize the anatomical composition of each thresholded map, suprathreshold voxels were assigned to AAL3v2 labels and then grouped into broader bilateral anatomical regions. This grouping was applied only after thresholding and did not affect the voxel weights or the optimization result.

The p90 threshold provided the best balance between reducing the voxel set and preserving the main anatomical pattern. At p90, the network contained 8,182 suprathreshold voxels. More liberal thresholds retained many more voxels (18,375 at p80 and 13,782 at p85), but did not reveal a substantially different anatomical pattern. All p90 anatomical groups were also present at p80 and p85, and the Jaccard overlap with the p90 region set remained high at p85 (0.95). In contrast, stricter thresholds began to remove regions: p95 retained 84.2% of the p90 groups, whereas p97.5 retained only 63.2%.

4.2 ROI-wise composition of the final network

The AAL3-derived atlas below was used to define ROIs in this study.

The final vigour network is also shown by broad anatomical sector below.

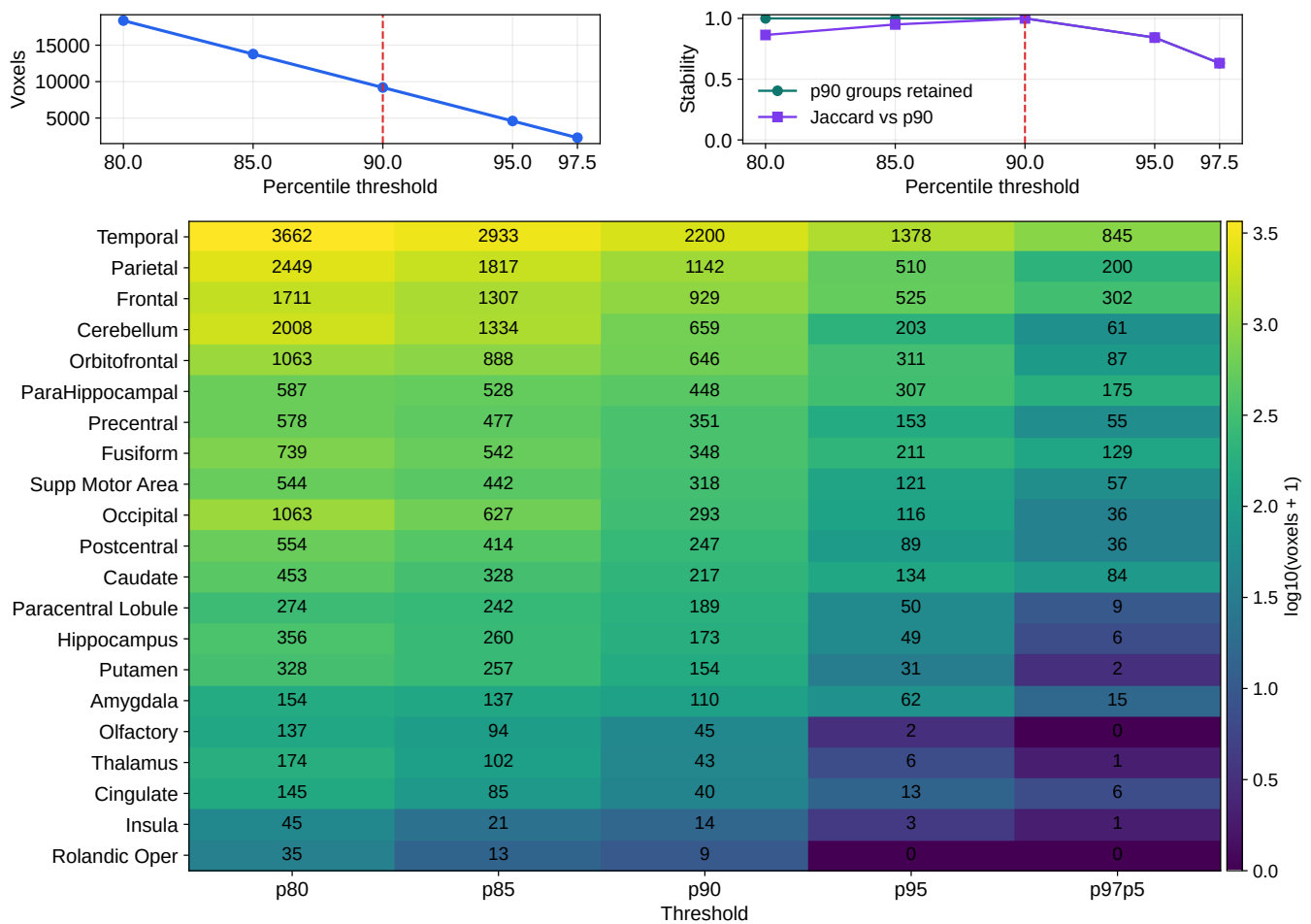


Figure S6: Threshold robustness analysis for the optimized vigour-related network. The voxel-weight threshold was varied across percentiles p80, p85, p90, p95, and p97.5. The panels show the number of retained voxels, overlap with the p90 region set, and anatomical composition of each thresholded map.

Table S3: ROI-wise composition of the suprathreshold vigour-related voxel-weight map shown in Fig. S8. The table uses the analysis-space NIfTI p90 source mask, which contains 8,182 selected voxels after excluding cerebral white matter and brainstem voxels. ROIs are AAL3-derived bilateral anatomical groups and are ranked by their contribution to the selected map. The listed ROI rows account for 100.00% of selected voxels.

ROI	ROI size	Selected voxels	% ROI selected	% of map
Temporal cortex	23811	2,068	8.69	25.27
Parietal cortex	18024	1,101	6.11	13.46
Frontal cortex	29358	931	3.17	11.38
Cerebellum	22855	646	2.83	7.90
Orbitofrontal cortex	4689	591	12.60	7.22
Fusiform gyrus	4445	364	8.19	4.45
Parahippocampal gyrus	1378	342	24.82	4.18
Precentral gyrus	5632	339	6.02	4.14
Supplementary motor area	3394	310	9.13	3.79
Occipital cortex	18157	278	1.53	3.40
Postcentral gyrus	6867	237	3.45	2.90
Caudate	1205	172	14.27	2.10
Paracentral lobule	1615	169	10.46	2.07
Putamen	1492	154	10.32	1.88

ROI	ROI size	Selected voxels	% ROI selected	% of map
Hippocampus	1088	116	10.66	1.42
Cerebral cortex	3610	113	3.13	1.38
Amygdala	411	109	26.52	1.33
Cingulate cortex	5668	38	0.67	0.46
Olfactory cortex	460	38	8.26	0.46
Thalamus	740	25	3.38	0.31
Nucleus accumbens	224	14	6.25	0.17
Insula	3529	13	0.37	0.16
Rolandic operculum	1892	9	0.48	0.11
Pallidum	523	5	0.96	0.06

After normalizing by ROI size, the strongest selection is concentrated in temporal-limbic and striatal regions, particularly the parahippocampal gyrus, amygdala, caudate, orbitofrontal cortex, and temporal cortex. Motor-related regions are also represented, including the supplementary motor area, paracentral lobule, precentral gyrus, parietal cortex, and cerebellum, but they do not dominate the ROI-normalized profile. Thus, the map contains a motor-related component, but its strongest ROI-level enrichment is distributed across temporal-limbic and associative regions rather than being confined to primary motor cortex.

4.3 Comparison with the task-engagement reference map

The tables below provide a compact summary of the limited overlap between the final vigour-related network and the task-engagement reference map, including both global overlap metrics and region-wise voxel counts.

Table S4. Comparison of the full-constraint vigour network and the task-activation reference map.

Network	Voxel count	Shared voxels (% of network)	Main anatomical labels
Task-activation map	21,610	835 (3.86%)	Cerebellum/vermis and thalamic nuclei; supramarginal cortex; occipital, parietal, and insular cortex
Full-constraint vigour network	8,182	835 (10.21%)	Parahippocampal cortex; amygdala; middle temporal pole; inferior temporal cortex; rectus and medial/posterior orbitofrontal cortex

ROI metrics: same-voxel overlap = 835 voxels; Dice coefficient = 0.056; Jaccard index = 0.029.

The shared voxels were concentrated most in putamen, cerebellar lobules 6/8, middle occipital cortex, paracentral lobule, superior parietal cortex, and postcentral gyrus.

In the GLM/GLMsingle comparison, the vigour network showed low overlap with each activation map: Dice = 0.09 with standard GLM, 0.11 with GLMsingle A, and 0.10 with GLMsingle D.

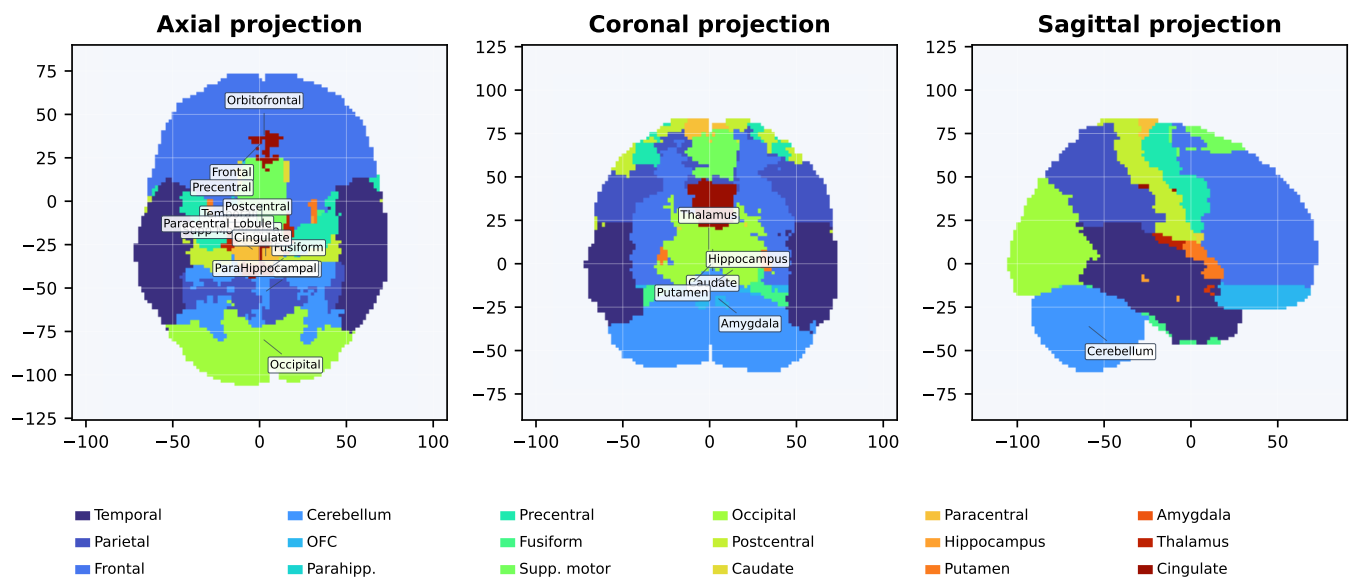


Figure S7: AAL3-derived bilateral ROI atlas used for anatomical summaries and ROI-level signal extraction. Axial, coronal, and sagittal projections show the grouped anatomical labels used in the network, medication-state, and connectivity analyses. Colors identify the broad bilateral ROI groups used for visualization and summary statistics.

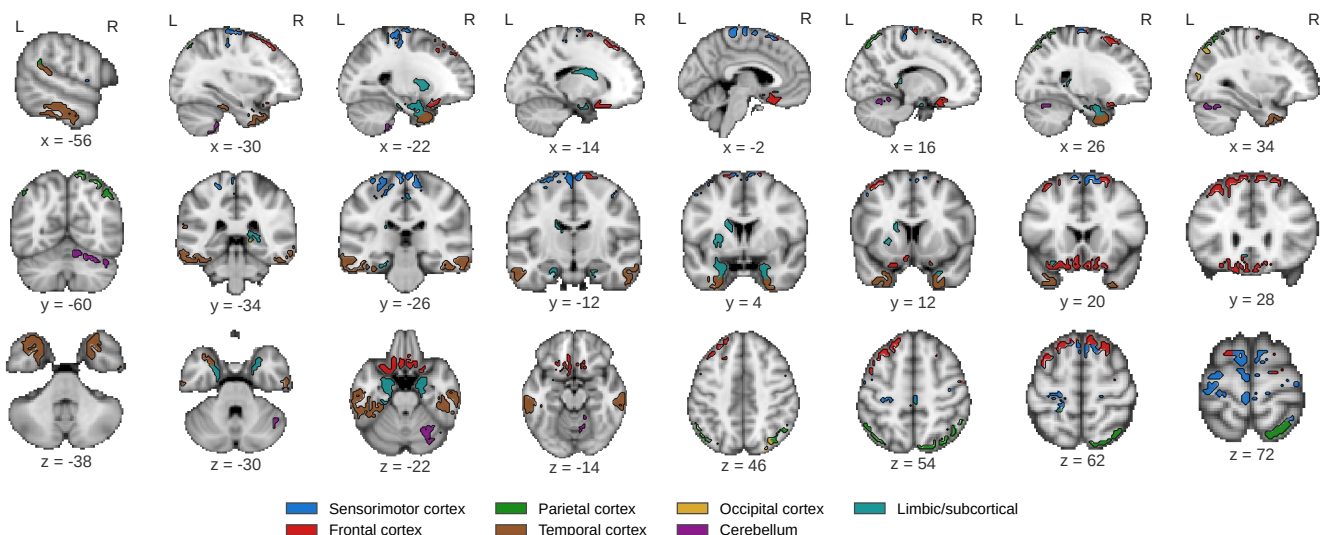


Figure S8: Anatomical-sector view of the optimized vigour-related network. The thresholded voxel-weight map is overlaid on representative sagittal, coronal, and axial MNI slices and colored by broad anatomical sector. The map includes sensorimotor, parietal, temporal, frontal, cerebellar, occipital, and limbic/subcortical components.

Table S5. ROI-level distribution of the vigour network, task-activation map, and their voxelwise overlap. The vigour network uses the NIfTI p90 source mask with 8,182 voxels; the task-activation map contains 21,610 voxels. ROI rows report total voxels in each map within each ROI, including shared voxels; these rows sum to 8,182 vigour voxels, 21,610 task-activation voxels, and 835 common voxels.

ROI	Shared voxel n (% ROI)	Vigour network n (% ROI)	Task activation n (% ROI)	Voxel-level relation
Putamen	118 (7.91%)	154 (10.32%)	710 (47.59%)	Shared voxels present
Pallidum	1 (0.19%)	5 (0.96%)	171 (32.70%)	Shared voxels present
Fusiform	28 (0.63%)	364 (8.19%)	979 (22.02%)	Shared voxels present
Parietal	181 (1.00%)	1,101 (6.11%)	4,352 (24.15%)	Shared voxels present
Postcentral	81 (1.18%)	237 (3.45%)	1,679 (24.45%)	Shared voxels present
Amygdala	0 (0.00%)	109 (26.52%)	0 (0.00%)	Vigour network only in ROI
Insula	0 (0.00%)	13 (0.37%)	908 (25.73%)	Both maps in ROI, no shared voxels
ParaHippocampal	0 (0.00%)	342 (24.82%)	5 (0.36%)	Both maps in ROI, no shared voxels
Occipital	86 (0.47%)	278 (1.53%)	3,538 (19.49%)	Shared voxels present
Cerebellum	233 (1.02%)	646 (2.83%)	4,259 (18.63%)	Shared voxels present
Temporal	43 (0.18%)	2,068 (8.69%)	2,015 (8.46%)	Shared voxels present
Paracentral_Lobule	19 (1.18%)	169 (10.46%)	122 (7.55%)	Shared voxels present
Caudate	0 (0.00%)	172 (14.27%)	0 (0.00%)	Vigour network only in ROI
Precentral	18 (0.32%)	339 (6.02%)	435 (7.72%)	Shared voxels present
Cingulate	5 (0.09%)	38 (0.67%)	724 (12.77%)	Shared voxels present
Hippocampus	0 (0.00%)	116 (10.66%)	29 (2.67%)	Both maps in ROI, no shared voxels
Rolandic_Oper	0 (0.00%)	9 (0.48%)	239 (12.63%)	Both maps in ROI, no shared voxels
Orbitofrontal	0 (0.00%)	591 (12.60%)	15 (0.32%)	Both maps in ROI, no shared voxels
Thalamus	0 (0.00%)	25 (3.38%)	57 (7.70%)	Both maps in ROI, no shared voxels
Supp_Motor_Area	2 (0.06%)	310 (9.13%)	43 (1.27%)	Shared voxels present
Olfactory	0 (0.00%)	38 (8.26%)	0 (0.00%)	Vigour network only in ROI
Frontal	19 (0.06%)	931 (3.17%)	1,258 (4.29%)	Shared voxels present
N_Acc	0 (0.00%)	14 (6.25%)	0 (0.00%)	Vigour network only in ROI
Cerebral_Cortex	1 (0.03%)	113 (3.13%)	72 (1.99%)	Shared voxels present

5 DETAILED EXPERIMENTAL AND PREPROCESSING METHODS

This section contains procedural details moved from the main manuscript to keep the main story readable while preserving reproducibility.

5.1 Full galvanic vestibular stimulation protocol

Galvanic vestibular stimulation (GVS) was delivered continuously throughout each run using an STMISOLA constant-current stimulator (BIOPAC Systems Inc.) through 41-mm Ag/AgCl electrodes placed bilaterally over the mastoid processes. Stimulation began before the first trial of the run. Electrode placement was verified using a 1 Hz sinusoidal stimulus, and stimulation amplitude was individualized using a staircase procedure to determine the highest current below the threshold for superficial cutaneous sensation at the electrode site (Nguyen et al., 2022). Participants were monitored throughout stimulation for discomfort or adverse effects.

Eight active amplitude-modulated waveforms were generated in MATLAB using suppressed-carrier modulation. The waveforms varied in carrier frequency (75 or 125 Hz) and envelope frequency (0, 10, 16, or 20 Hz). The resulting GVS stimuli were labeled as follows: GVS1, 75 Hz carrier and 0 Hz envelope; GVS2, 75 Hz carrier and 10 Hz envelope; GVS3, 75 Hz carrier and 16 Hz envelope; GVS4, 75 Hz carrier and 20 Hz envelope; GVS5, 125 Hz carrier and 0 Hz envelope; GVS6, 125 Hz carrier and 10 Hz envelope; GVS7, 125 Hz carrier and 16 Hz envelope; and GVS8, 125 Hz carrier and 20 Hz envelope. A sham

condition, in which no current was delivered (0 mA), was also included. These frequencies were selected to approximate typical vestibular afferent firing rates and oscillatory bands implicated in motor planning and execution.

Each run was divided into nine blocks, with every block corresponding to a single stimulation condition: one of the eight active GVS waveforms or the sham condition. Each block contained 10 trials, and the assigned stimulation condition was active throughout the full block, including the cue, delay, Go-signal, and response periods. All nine conditions were presented exactly once per run. At block transitions, the next stimulation condition began immediately, except after every third block, when a 20 s rest period was provided before the next condition. The order of the nine blocks was randomized independently for each run, so that the sequence of conditions differed across runs, sessions, and subjects. Participants were not informed of the stimulation order. Because stimulation conditions were presented in a blocked sequence without a dedicated washout interval between all adjacent blocks, carryover effects between successive stimulation conditions cannot be fully excluded. Randomization of block order was used to reduce systematic order effects, and GVS-related results were therefore interpreted cautiously.

5.2 Full MRI acquisition and preprocessing protocol

Structural and functional MRI data were acquired on a Philips 3T Ingenia Elition X whole-body MRI scanner upgraded to an MR7700 system (Philips Healthcare, Best, The Netherlands) using a 32-channel head coil. For each participant, a T1-weighted structural image was acquired, followed by two task-based fMRI scans, one for each run of the behavioural task.

T1-weighted images were acquired using a 3D MPRAGE sequence with the following parameters: flip angle = 8° , echo time (TE) = 3.6 ms, repetition time/shot interval (TR) = 2400 ms, inversion time = 950 ms, 180 slices, field of view (FOV) = $256 \times 256 \times 180 \text{ mm}^3$, and spatial resolution = 1 mm^3 isotropic. Images were acquired with a compressed SENSE factor of 3.6, and the acquisition time was 3:42 min.

Functional MRI data were acquired using echo-planar imaging (EPI). The functional acquisition protocol differed for the first three participants compared with the rest of the sample. The first three participants were scanned with a multi-echo EPI protocol, with flip angle = 77° , TE = 12 ms, TR = 2000 ms, 48 slices, FOV = $240 \times 144 \times 240 \text{ mm}^3$, spatial resolution = 3 mm^3 isotropic, SENSE factor = 2, multiband factor = 2, and acquisition time = 14:22 min. The remaining participants were scanned with a single-echo EPI protocol, with flip angle = 77° , TE = 30 ms, TR = 1000 ms, 51 slices, FOV = $240 \times 153 \times 240 \text{ mm}^3$, spatial resolution = 3 mm^3 isotropic, SENSE factor = 2, multiband factor = 3, and acquisition time = 14:17 min. To support distortion correction, two spin-echo EPI images were acquired with opposite fat-shift directions, anterior–posterior and posterior–anterior. These spin-echo images were acquired with flip angle = 77° , TE = 55 ms, TR = 2650 ms, 51 slices, FOV = $240 \times 153 \times 240 \text{ mm}^3$, spatial resolution = 3 mm^3 isotropic, SENSE factor = 2, multiband factor = 3, and acquisition time = 5.3 s.

Image preprocessing was performed in FSL (Smith et al., 2004). Structural images underwent bias-field correction and brain extraction. Functional images were corrected for head motion and susceptibility-induced distortions, with distortion correction referring to the correction of susceptibility-related spatial distortions in the functional images using the spin-echo images acquired with opposite fat-shift directions. No initial functional volumes were discarded during preprocessing because the scanner automatically discarded four dummy volumes before the start of saved image acquisition. Functional images were then high-pass filtered using a 100 s cutoff, brain extracted, denoised using ICA-AROMA (Pruim et al., 2015), and spatially smoothed with a 5 mm full-width at half-maximum Gaussian kernel. The preprocessed functional images were then registered to each participant's structural image and normalized to standard MNI152 space.

5.3 Standard GLM task-activation reference

To provide a brain map of task-evoked activation, we first ran a standard GLM using FSL FEAT version 6.00 (Jenkinson et al., 2012). This analysis was separate from the trial-wise GLM described below and was used only to generate the standard task-activation map shown in Supplementary Figure S4.

For each run, all valid Go trials (Supplementary Figure S1) were modeled with one event regressor. Event onsets were defined as the Go-signal times, with a duration of 2 s and an amplitude of 1. This regressor

was convolved with FSL's double-gamma hemodynamic response function, and a temporal derivative was included.

The first-level contrast tested the positive effect of task activity. The two runs from each subject-session were then combined using a fixed-effects model. The resulting subject-session maps were entered into a higher-level mixed-effects model using FLAME stage 1 (Woolrich et al., 2004; Beckmann et al., 2003). The group model included one mean regressor and tested whether the average task response was greater than zero across the 35 available subject-sessions. Group Z-statistic maps were thresholded using clusters formed at $Z \geq 3.1$ with a corrected cluster significance threshold of $p \leq 0.05$ (Worsley, 2001). The thresholded group map was used as the standard task-activation map. (Supplementary Figure S4)

5.4 Trial-wise GLM estimation and beta preprocessing details

Task-related BOLD responses for each trial were estimated using trial-wise general linear model (GLM) beta estimates with GLMsingle (Kay et al., 2013; Prince et al., 2022). Analyses were performed separately for each subject-session. Within each subject-session, the two task runs were analyzed jointly, while run identity was preserved in the design matrix and nuisance regressors.

The GLM design matrix included Go-signal onset regressors with a fixed trial duration of 2 s, corresponding to the period during which participants performed the task. Catch trials were not modeled as separate regressors and were excluded from the analysis because participants did not squeeze the bulb on these trials (630 of 6,300 trials across all datasets). Reward-condition labels were also not included in the GLM design. To restrict the analysis to relevant tissue, background, cerebrospinal fluid (CSF), and white-matter voxels were masked out, retaining only gray-matter voxels. The initial MNI gray-matter, non-CSF candidate mask contained 91,781 voxels. The mean CSF time series for each run was included as a nuisance regressor.

GLMsingle was then run with HRF fitting, GLMdenoise, and fractional ridge regression enabled. HRF fitting used the default GLMsingle library of 20 candidate HRFs, and the HRF yielding the highest variance explained was selected separately for each voxel. GLMdenoise derived additional nuisance regressors from low task-related-variance voxels using principal components analysis and selected the number of noise components by cross-validation. Fractional ridge regression was used to stabilize single-trial beta estimates. Shrinkage fractions from 0.05 to 1.00 in increments of 0.05 were evaluated, and the optimal shrinkage level was selected separately for each voxel by cross-validation. Run-specific baseline terms were modeled using GLMsingle's run-wise polynomial regressors, including degree-0 intercepts and higher-order polynomial terms up to degree $\text{round}(L/2)$ for a run of length L minutes.

A second cleaning step was then applied to the GLMsingle beta estimates. To detect outlier beta values, beta estimates were robustly normalized within each subject-session using the session-level median and MAD:

$$\beta_{\text{norm}} = \frac{\beta - \text{median}(\beta)}{\text{MAD}(\beta)}.$$

Beta values with $|\beta_{\text{norm}}|$ above the 99.9th percentile were flagged as beta outliers. For each voxel, the beta-outlier fraction was computed as the proportion of trials flagged as outliers. Voxels were retained when this fraction was ≤ 0.5 and excluded when more than 50% of their trials were beta outliers. This majority-trial threshold was used to remove voxels with persistently unstable beta estimates, rather than voxels with occasional large responses.

This cleaning retained $91,701 \pm 54$ voxels on average. This corresponded to an average removal of $0.09\% \pm 0.06\%$. To assess whether this filtering disproportionately affected susceptibility-prone ventral regions, removal rates were summarized using atlas labels and ventral-position proxies. Average run-level removal was 0.20% in frontal orbital cortex, 0.06% in temporal pole, 0.76% in inferior temporal gyrus, 0.25% in the ventral bottom quartile, and 0.03% in all other voxels. Thus, run-level cleaning removed very few voxels overall; removal was somewhat higher in inferior temporal and ventral-bottom voxels, but absolute removal rates remained below 1% in all regions.

The cleaned outputs from all subjects, sessions, and runs were then used to define a shared voxel space for group-level analysis. The coverage threshold was applied across all concatenated trial observations

from all subjects, sessions, and runs, rather than separately within each subject or session. This approach was used because the group-level analysis required a common MNI-space voxel feature set across trials. Applying separate subject-level masks would have produced different voxel sets across subjects, whereas the shared mask allowed all valid trial-wise beta estimates to be aligned into one group-level matrix. The threshold was also permissive: voxels were retained if they had finite beta estimates for at least 70% of all trial observations, so a voxel was not excluded simply because it was missing or unstable in a minority of subjects, sessions, or runs.

The final group coverage mask contained 91,769 voxels, yielding a final group beta matrix of 91,769 voxels $\times$ 5,463 retained trials (Supplementary Figure S1). This mask was generated using a 70% finite-coverage criterion over the source trial slots, not by intersecting voxels across all runs. The filter excluded 12 voxels from the 91,781-voxel candidate mask, retaining 99.99% of candidate voxels. Subsequent analyses used the concatenated cleaned beta data in this final group voxel space.

6 FULL OPTIMIZATION DERIVATION AND MODEL-SELECTION DETAILS

A group-level optimization procedure was used to identify a vigour-related brain network (main Figure 1B,C). The aim of the optimization framework was to identify a voxel-weight vector that produced a projected neural signal related to trial-wise motor vigour while satisfying constraints on task engagement, spatial coherence, and low consecutive-trial variability. In voxel space, this projection is defined by a voxel-weight vector, $\mathbf{w}^{\text{org}}$, that combines voxel activity into one scalar signal at each sample. For a neural data matrix $\mathbf{X}^{\text{org}} \in \mathbb{R}^{N_v \times N_s}$, where columns are samples, the projected signal is

$$\mathbf{y}^{\text{org}} = (\mathbf{w}^{\text{org}})^{\top} \mathbf{X}^{\text{org}}, \quad \mathbf{w}^{\text{org}} \in \mathbb{R}^{N_v \times 1}, \mathbf{y}^{\text{org}} \in \mathbb{R}^{1 \times N_s}.$$

Here, N_v is the number of voxels and N_s denotes the time sample of the neural data matrix. In this study, the same projection can be applied to two forms of neural data: single-trial beta estimates and trial-window BOLD time courses:

$$\mathbf{y}_{\beta}^{\text{org}} = (\mathbf{w}^{\text{org}})^{\top} \mathbf{X}_{\beta}^{\text{org}}, \quad \mathbf{X}_{\beta}^{\text{org}} \in \mathbb{R}^{N_v \times N_{\text{tr}}}, \mathbf{y}_{\beta}^{\text{org}} \in \mathbb{R}^{1 \times N_{\text{tr}}}.$$

$$\mathbf{y}_{\text{BOLD}}^{\text{org}} = (\mathbf{w}^{\text{org}})^{\top} \mathbf{X}_{\text{BOLD}}^{\text{org}}, \quad \mathbf{X}_{\text{BOLD}}^{\text{org}} \in \mathbb{R}^{N_v \times N_{\text{tp}}}, \mathbf{y}_{\text{BOLD}}^{\text{org}} \in \mathbb{R}^{1 \times N_{\text{tp}}}.$$

Here, N_{tr} is the number of trials, and N_{tp} is the number of time samples. $\mathbf{X}_{\beta}^{\text{org}}$ and $\mathbf{X}_{\text{BOLD}}^{\text{org}}$ were not separate models; they were two representations of the neural data used by the same projection framework.

Although the framework is defined in voxel space, direct optimization over all voxels was impractical because of the high dimensionality of the data and memory limitations. In addition, the task was not a simple motor task and included several task-related and non-motor processes. Therefore, reducing components that were unlikely to reflect the motor-vigour-related signal was important before optimization. Because missing values remained after beta-estimate preprocessing, the neural data were reduced using expectation-maximization principal component analysis (EMPCA) (Roweis, 1997). A 80% cumulative explained-variance threshold initially retained 14 components.

EMPCA provided voxel-to-PC weights, $\mathbf{W}_{\text{pca}} \in \mathbb{R}^{N_v \times N_{\text{pc}}}$. Both neural data matrices were projected into this common PCA space:

$$\begin{aligned} \mathbf{X}_{\beta} &= \mathbf{W}_{\text{pca}}^{\top} \mathbf{X}_{\beta}^{\text{org}}, & \mathbf{X}_{\beta} &\in \mathbb{R}^{N_{\text{pc}} \times N_{\text{tr}}}, \\ \mathbf{X}_{\text{BOLD}} &= \mathbf{W}_{\text{pca}}^{\top} \mathbf{X}_{\text{BOLD}}^{\text{org}}, & \mathbf{X}_{\text{BOLD}} &\in \mathbb{R}^{N_{\text{pc}} \times N_{\text{tp}}}. \end{aligned}$$

The optimization was then performed in PCA space using $\mathbf{w} \in \mathbb{R}^{N_{\text{pc}} \times 1}$. In this space, the projected beta and BOLD signals were

$$\begin{aligned} y_\beta &= \mathbf{w}^\top \mathbf{X}_\beta, & y_\beta &\in \mathbb{R}^{1 \times N_{\text{tr}}}, \\ y_{\text{BOLD}} &= \mathbf{w}^\top \mathbf{X}_{\text{BOLD}}, & y_{\text{BOLD}} &\in \mathbb{R}^{1 \times N_{\text{tp}}}. \end{aligned}$$

The total loss was defined using these two projected signals: the beta projection was used for the trial-wise beta-based terms, including behavioural relevance to motor vigour, and the BOLD projection was used for the consecutive-trial variability term. After optimization, the PCA-space weights were projected back to voxel space as $\mathbf{w}^{\text{org}} = \mathbf{W}_{\text{pca}} \mathbf{w}$. The resulting voxel-space weight vector is considered to define the group-level vigour-related network.

The optimization model was based on four assumptions about the neural representation of motor vigour. First, the target network should consist of voxels engaged during task performance. Second, the projected signal should encode behavioural information. Third, neighbouring voxels should show similar contributions, reflecting spatial coherence. Fourth, the selected signal should show low consecutive-trial variability. These four conditions were incorporated into the model one by one, as described below.

Condition 1: The projected signal should come from task-engaged voxels.

The first condition was that the projected signal should be generated mainly from voxels that were engaged during task performance. Each voxel contributes to y_β through its beta activity and its corresponding weight in w . Therefore, if a voxel has weak task-related beta activation, assigning a small weight to that voxel would allow a high task-engaged region to contribute strongly to the projected signal:

$$\min_w \sum_j \frac{w_j^2}{|X_{\beta,j}|} = \min_w w^\top \text{diag}\left(\frac{1}{|X_\beta|}\right) w.$$

This gives the task-engagement loss term:

$$L_{\text{task}}(w) = w^\top C_{\text{task}} w, \quad C_{\text{task}} = \text{diag}\left(\frac{1}{|X_\beta|}\right).$$

Thus, C_{task} favours voxels with stronger task-related beta responses and discourages the model from selecting voxels with weak task engagement.

Condition 2: The projected signal should encode behavioural information.

The second condition was that the projected neural signal should be behaviourally relevant. In this study, the target behavioural variable was reaction time. Therefore, the model should find w such that y is strongly associated with the behavioural variable X_{behav} :

$$\begin{aligned} \max_w \text{Corr}(X_{\text{behav}}, y_\beta)^2 &= \max_w \text{Corr}(X_{\text{behav}}, w^\top X_\beta)^2 \\ &= \max_w \frac{[(w^\top X_\beta) X_{\text{behav}}^\top]^2}{w^\top (X_\beta X_\beta^\top) w} \\ &= \max_w \frac{w^\top (X_\beta X_{\text{behav}}^\top X_{\text{behav}} X_\beta^\top) w}{w^\top (X_\beta X_\beta^\top) w} \\ &= \max_w \frac{w^\top C_b w}{w^\top C_d w}, \end{aligned}$$

where

$$C_b = X_\beta X_{\text{behav}}^\top X_{\text{behav}} X_\beta^\top, \quad C_d = X_\beta X_\beta^\top.$$

Here, C_b captures the component of the projected beta signal that is aligned with behaviour, while C_d normalizes the association by the variance of the projected beta signal. Since the final optimization problem is written as a minimization problem, this behavioural-relevance term is later included with a negative sign.

Condition 3: The projected signal should come from a spatially coherent network.

The third condition was that the selected pattern should form a spatially coherent network rather than a scattered set of isolated voxels. It means neighbouring voxels were encouraged to have similar weights. If E denotes the set of neighbouring voxel pairs, this penalty can be written as

$$\min_w \sum_{(p,q) \in E} (w_p - w_q)^2. \quad (\text{S1})$$

A spatial adjacency matrix was constructed from voxel anatomy, and its graph Laplacian was used to define the spatial smoothness penalty. This gives

$$L_{\text{smooth}}(w) = w^\top C_{\text{smooth}} w, \quad C_{\text{smooth}} = L_{\text{graph}}. \quad (\text{S2})$$

Thus, C_{smooth} enforces spatial smoothness across neighbouring voxels. When the optimization is performed in PCA space, this spatial smoothness penalty is expressed in the same reduced space before optimization.

Condition 4: The projected signal should show low consecutive-trial variability.

The fourth condition was that the selected signal should exhibit low consecutive-trial variability. Variability was therefore defined as the difference between consecutive trials, rather than as variance across the full time series. This distinction is important because variance captures the overall dispersion of a signal around its mean but does not account for trial order. As a result, two projected signals may have similar variance even if one evolves smoothly across trials while the other fluctuates sharply from one trial to the next. Consecutive-trial differences, in contrast, preserve the local temporal structure of the signal and directly measure abrupt changes between neighbouring trials.

This definition is motivated by the behavioural structure of motor vigour. Vigour on trial i is unlikely to be statistically independent from vigour on trial $i + 1$, because consecutive trials are shaped by similar physiological, attentional, and task-related states. Trials that occur close in time are therefore expected to be more similar than trials separated by longer intervals. Accordingly, a stable vigour-related neural signal should not contain large unexplained jumps between consecutive trials. This reasoning is consistent with previous work on reaction-time variability showing that standard deviation alone does not capture temporal dependency, whereas measures such as lag-one autocorrelation and successive-trial differences reveal the local structure of performance fluctuations (Perquin et al., 2023).

A similar principle has been applied in fMRI research using mean squared successive difference (MSSD) and related measures, which quantify moment-to-moment changes in BOLD activity rather than dispersion around a global mean (Garrett et al., 2011; Wehrheim et al., 2024; Nomi et al., 2018). This approach is particularly appropriate for task fMRI, where the expected signal may change gradually over time, across task conditions, or between blocks. A global variance measure could incorrectly treat such slow task-related changes as variability, whereas a consecutive-difference penalty specifically targets local instability between consecutive trials. Thus, the low-variability term in the present model penalized large consecutive-trial changes in the projected signal while allowing slower task-related changes to remain. Boundaries between runs and blocks were excluded from this penalty because trials across these boundaries do not reflect continuous local temporal dynamics and may contain artificial discontinuities introduced by changes in run structure or task context.

We need to minimize variability between consecutive trials:

$$\begin{aligned}
\min_w \sum_{i \in I_c} \|y_{\text{bold},i} - y_{\text{bold},i+1}\|_2^2 &= \min_w \sum_{i \in I_c} (y_{\text{bold},i} - y_{\text{bold},i+1}) (y_{\text{bold},i} - y_{\text{bold},i+1})^\top \\
&= \min_w \sum_{i \in I_c} (w^\top X_{\text{bold},i} - w^\top X_{\text{bold},i+1}) (w^\top X_{\text{bold},i} - w^\top X_{\text{bold},i+1})^\top \\
&= \min_w \sum_{i \in I_c} w^\top (X_{\text{bold},i} - X_{\text{bold},i+1}) (X_{\text{bold},i} - X_{\text{bold},i+1})^\top w. \quad (\text{S3})
\end{aligned}$$

Here, I_c denotes the set of valid consecutive-trial pairs after excluding run and block boundaries. This gives the BOLD variability loss:

$$\begin{aligned}
L_{\text{bold}}(w) &= w^\top C_{\text{bold}} w, \\
C_{\text{bold}} &= \sum_{i \in I_c} (X_{\text{bold},i} - X_{\text{bold},i+1}) (X_{\text{bold},i} - X_{\text{bold},i+1})^\top. \quad (\text{S4})
\end{aligned}$$

Therefore, C_{bold} penalizes large changes in BOLD activity between consecutive trials.

The same low-variability condition was also applied to the beta estimates:

$$\begin{aligned}
\min_w \sum_{i \in I_c} \|y_{\beta,i} - y_{\beta,i+1}\|_2^2 &= \min_w \sum_{i \in I_c} (y_{\beta,i} - y_{\beta,i+1}) (y_{\beta,i} - y_{\beta,i+1})^\top \\
&= \min_w \sum_{i \in I_c} (w^\top X_{\beta,i} - w^\top X_{\beta,i+1}) (w^\top X_{\beta,i} - w^\top X_{\beta,i+1})^\top \\
&= \min_w \sum_{i \in I_c} w^\top (X_{\beta,i} - X_{\beta,i+1}) (X_{\beta,i} - X_{\beta,i+1})^\top w. \quad (\text{S5})
\end{aligned}$$

This gives the beta variability loss:

$$\begin{aligned}
L_{\beta}(w) &= w^\top C_{\beta} w, \\
C_{\beta} &= \sum_{i \in I_c} (X_{\beta,i} - X_{\beta,i+1}) (X_{\beta,i} - X_{\beta,i+1})^\top. \quad (\text{S6})
\end{aligned}$$

Therefore, C_{β} penalizes large changes in beta values between consecutive trials.

Both BOLD and beta variability terms were included because they capture complementary sources of variability. The beta estimates reflect activation related to movement execution, so $L_{\beta}(w)$ constrains variability in the movement-related neural response. However, the task also includes other processes. These sources of variability can be encoded in the BOLD signal even when they are not fully captured by movement-related beta estimates. Therefore, $L_{\text{bold}}(w)$ was also included to account for broader task-related variability encoded in the brain.

General loss function: After defining the four conditions, the penalty terms were combined into a single total loss. The total penalty was defined as

$$L_{\text{total}}(w) = \alpha_1 L_{\text{task}}(w) + \alpha_2 L_{\text{bold}}(w) + \alpha_3 L_{\beta}(w) + \alpha_4 L_{\text{smooth}}(w) = w^{\top} C_{\text{total}} w,$$

where

$$C_{\text{total}} = \alpha_1 C_{\text{task}} + \alpha_2 C_{\text{bold}} + \alpha_3 C_{\beta} + \alpha_4 C_{\text{smooth}}. \quad (\text{S7})$$

The parameters α_1 , α_2 , α_3 , and α_4 control the relative contributions of task engagement, BOLD variability, beta variability, and spatial smoothness, respectively.

The final objective combined this total penalty with behavioural relevance. The model minimized the regularization penalties while maximizing the association between the projected beta signal and reaction time:

$$\min_w \frac{\gamma w^{\top} C_{\text{total}} w - w^{\top} C_b w}{w^{\top} C_d w} \quad \text{s.t. } \|w\|_2 = 1. \quad (\text{S8})$$

Here, γ controls the balance between satisfying the regularization criteria and maximizing behavioural relevance. The denominator $w^{\top} C_d w$ normalizes the objective by the variance of the projected beta signal.

The resulting projection was defined as

$$y = w^{\top} X_{\beta}. \quad (\text{S9})$$

The optimization problem was solved under the unit-norm constraint on w using the Sequential Least Squares Programming (SLSQP) algorithm (Nocedal and Wright, 2006; Kraft, 1988). The objective combined a behavioural relevance term with four regularization terms. The coefficients α_{task} , α_{BOLD} , α_{β} , and α_{smooth} controlled the relative contribution of the task-engagement, BOLD trial-to-trial variability, beta trial-to-trial variability, and spatial smoothness penalties, respectively. The scalar γ controlled the overall strength of the combined penalty terms relative to the behavioural relevance term. Hyperparameters were selected using a coarse-to-fine cross-validation procedure. In the coarse stage, the model was evaluated over a predefined set of candidate penalty-weight combinations chosen to span weaker, balanced, and stronger weighting of the different penalties. Trials were divided into contiguous folds, and for each candidate setting the model was fit on the training folds and evaluated on the held-out fold using the absolute correlation between the optimized neural projection and behaviour. After the best-performing coarse setting was identified, a fine search was performed by testing small multiplicative perturbations around that setting, both for individual penalty weights and for the overall balance of all penalty weights. The final hyperparameter setting was chosen based on held-out correlation, with preference given to settings that showed stable performance across folds and small train-test differences. For each fold, the optimization was then solved independently using the training data, and the resulting weights were projected back from PCA space to voxel space. Final voxel maps and projected signals were obtained by averaging across folds. The resulting weight assigned to each voxel reflected its contribution to satisfying the combined criteria of task engagement, behavioural relevance, spatial smoothness, and low trial-to-trial variability. Voxels with the largest weights were then defined as the target network.

Finally, we defined the vigour network as voxels whose weights exceeded the 90th percentile. The reason we chose this threshold is that the 90th percentile provided a conservative but stable compromise between spatial specificity and anatomical coverage (Supplementary Figure S6). Lower thresholds, such as the 80th or 85th percentiles, retained substantially more voxels and therefore produced a broader, less selective network. In contrast, more stringent thresholds, particularly the 95th and 97.5th percentiles, sharply reduced the number of retained voxels and began to exclude several anatomical regions. The 90th percentile retained a distributed network across the main contributing regions while showing maximal stability relative to the reference threshold, supporting it as an appropriate cutoff for defining the vigour network.

After identifying the vigour network, we then assessed whether medication state or GVS condition modulated activity within this network, as both variables had been excluded from the optimization procedure.

7 MEDICATION-STATE CONTROL ANALYSES

7.1 Control analysis: no evidence for preferential ON-state stabilization

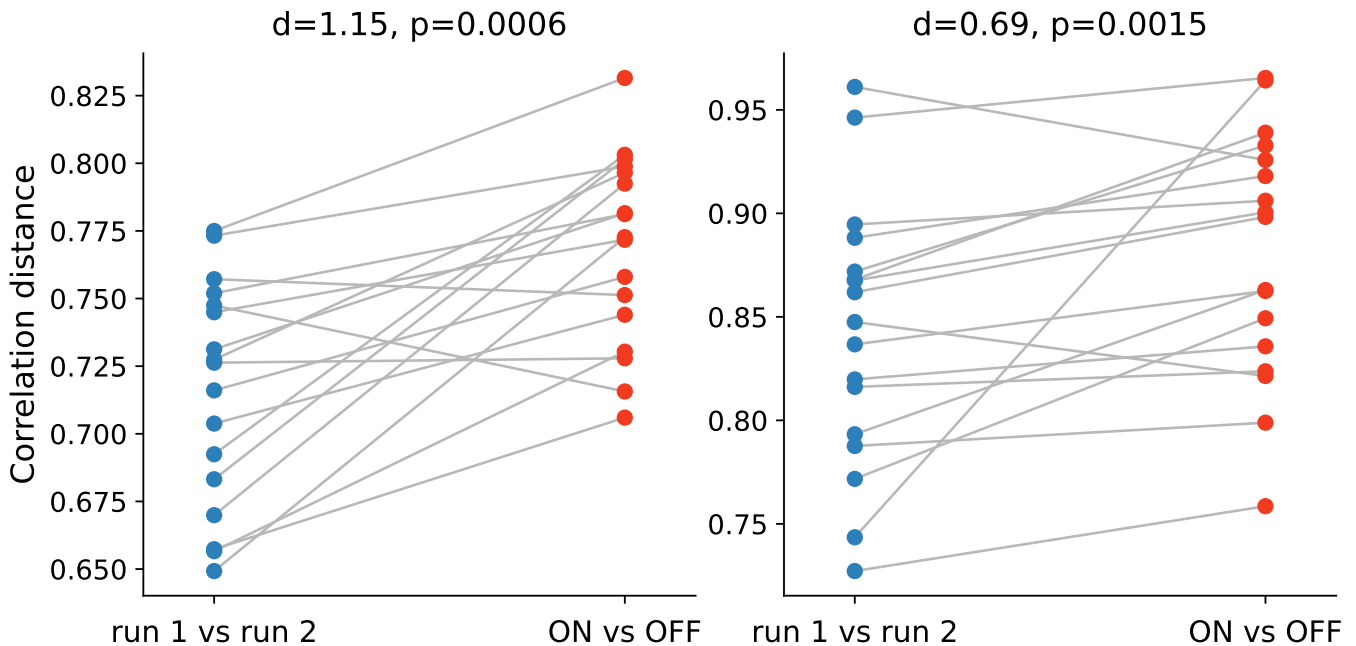


Figure S9: Medication-state control analysis for voxel-pattern stability. Subject-level paired correlation distances compare within-state run 1 versus run 2 distances with ON- versus OFF-medication distances. Lines connect paired estimates from the same participant. The ON-versus-OFF distance exceeded run-to-run distance in both panels, indicating that medication state changed voxel-pattern structure relative to within-state reliability but did not provide evidence for preferential ON-state stabilization of the selected voxels.

8 GVS-RELATED SUPPLEMENTARY ANALYSES

The main manuscript reports the mutual-information GVS edge-change analysis in both the optimized vigour network and the task-activation network. The supplementary figure below shows the corresponding Spearman-rank analysis for the optimized vigour network. Spearman correlation was included because it measures monotonic rank-based association and is less sensitive to extreme beta values than Pearson correlation. As in the main analysis, active-minus-sham edge changes were pooled across subjects, medication states, runs, and active GVS waveforms, tested using sign-flip permutation, and corrected with FDR within the metric and analysis scope.

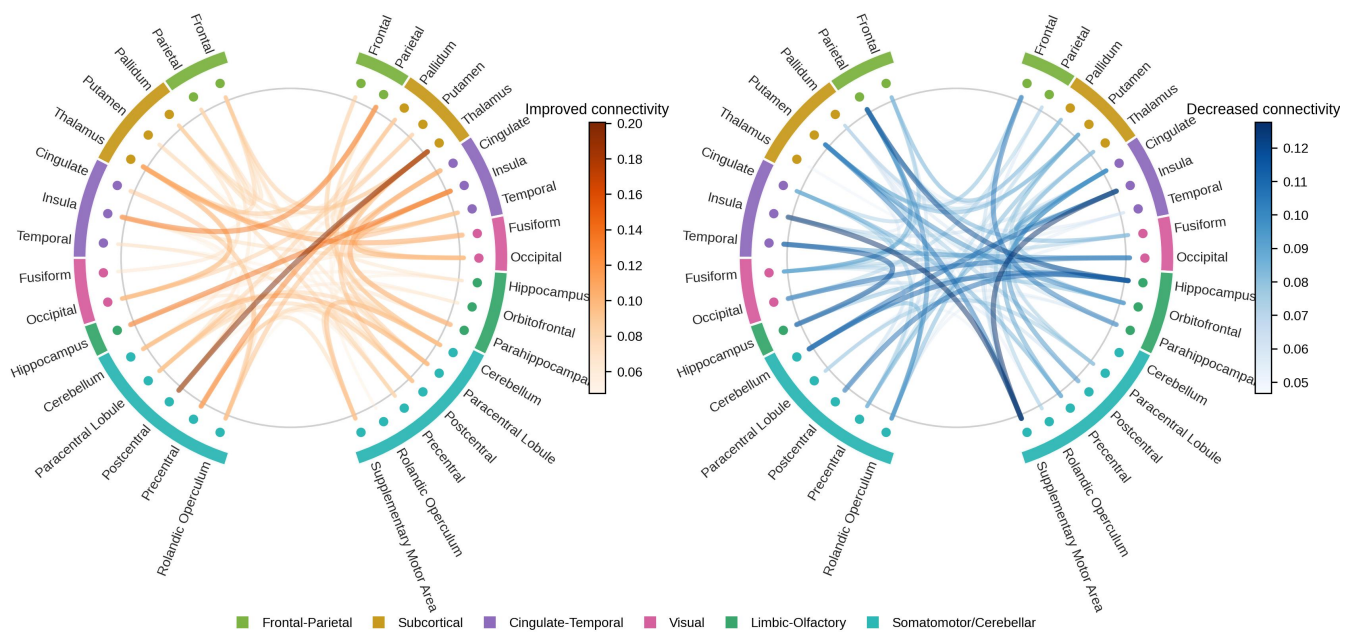


Figure S10: Exploratory GVS-related Spearman edge changes in the optimized vigour network. FDR-significant active-minus-sham Spearman edge changes are shown after pooling across subjects, medication states, runs, and active GVS waveforms. Orange edges indicate increased rank-based association and blue edges indicate decreased association; color intensity scales with the mean edge-change magnitude. Nodes are labeled by ROI and grouped by anatomical sector.

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